

breaches, and misuse are operational risks that are affecting consumers, enterprises, and public institutions. The line between digital harm and business harm has effectively disappeared.

The growing trust deficit

AI models continue to struggle with hallucinations and inconsistent predictions.

Healthcare provides some of the clearest evidence. A model incorrectly predicting diabetes for an individual led them to start the wrong medication, with serious consequences. Another model offered flawed medical suggestions that understandably alarmed users. When systems behave this way, trust erodes faster than technical teams can explain what went wrong.

The environmental impact adds another dimension. A single ChatGPT-style prompt produces about 4.32 grams of CO₂. That may seem minor, but at enterprise scale, it adds up fast. When millions of queries flow through models running in non-optimised data centres, the emissions rise sharply.

Operating costs also directly correlate with model maintenance and can increase with scalability challenges. The need to maintain constant accuracy in the face of drift and other risks further complicates responsible deployment.

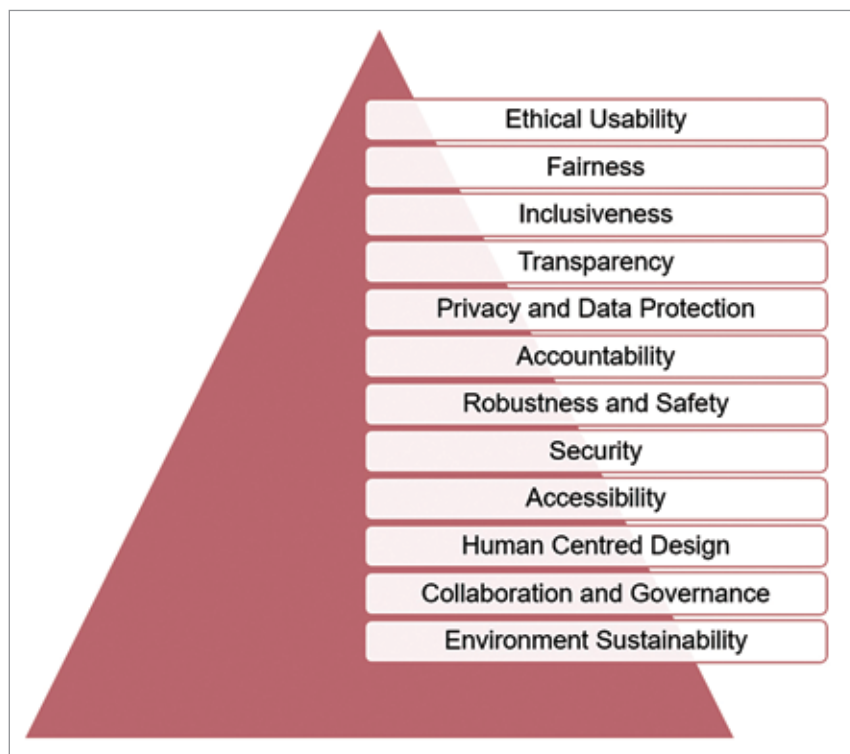
Principles of responsible AI

Principles of responsible AI ensure that AI continues to work equitably across situations and does not dilute results based on users' different characteristics. Here is a quick look at the principles that I recommend.

Ethical usability: AI applications should be created only for ethical usage.

Fairness: This is where it all starts. AI must treat everyone fairly. It should not favour one group over another. For example, an AI used for hiring should not prefer one gender or community.

Transparency: We must understand how AI arrives at critical decisions. In this regard, explainability tools such



Gartner's PRISM model for principles of responsible AI

as SHAP and LIME exist. Their core function is to explain why a model made a particular choice. They use model cards and training summaries to show the entire workflow by which the AI derived its result.

Privacy: Users should be aware of which of their data is being collected by AI and how that data is used. They can only be in control when they have informed consent and the right to delete any data they are uncomfortable sharing.

Accountability: Every AI system should have a structured reporting and grievance-resolution process with a clearly accountable team or individual. Users should be able to report issues and have them fixed.

Safety: AI must operate reliably even when conditions change. Systems should be tested to handle unexpected situations or manipulations. For example, a chatbot should not provide incorrect advice if someone attempts to confuse it intentionally. Failure analysis and real-world testing help make AI dependable.

Security: AI must be protected against cyberattacks. Strong measures should be in place to prevent misuse and data poisoning.

Accessibility: Disabilities and/or inadequate technical knowledge should not prevent any users from accessing and safely using AI. A simple interface and an inclusive user experience can be effective.

Sustainability: Every AI operation should adopt more sustainable practices to remain consistently energy-efficient. These can include algorithm optimisation and the establishment of eco-friendly data centres.

Collaboration: Cross-functional teams can help implement the principles outlined above more effectively.

How mandatory compliance is evolving with regulatory pressures

Regulation has finally caught up with technological scale. Industry research indicates that more than 99 per cent of